Circles

Lesson 1

Teacher's copy: overview.

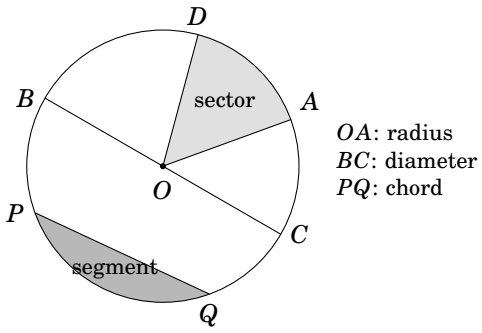
- **Aims.** Students use the terms of a circle correctly. They explain why equal chords subtend equal angles at the centre, and the converse. They explain why the perpendicular from the centre bisects a chord, and the converse. They find chord lengths and distances with the Pythagoras theorem.
- **Syllabus.** CBSE Class IX 2026–27, Unit IV Geometry (25 of 80 marks). The syllabus asks students to *explain why* each result holds. The explanations here use congruence (SSS, SAS, RHS).
- **Method.** Students discover each result by solving a short problem first. Give 3–4 minutes, then discuss the answers and state the theorem. Students need a ruler and a compass.
- **Timing.** Section 1: 5 min. Section 2: 15 min. Section 3: 20 min. Practice: 15 min. Exit question: 5 min.
- **Next lesson.** The circle through three points and the circumcircle of a triangle.

1 Parts of a circle (5 min)

Start with familiar circles: the wheel of a bullock cart and the Ashoka Chakra on the national flag. Ask: what is the same about every spoke of a wheel?

Definition 1. A **circle** is the set of all points in a plane at a fixed distance from a fixed point. The fixed point is the **centre**. The fixed distance is the **radius**.

- A **chord** is a line segment that joins two points on the circle.
- A **diameter** is a chord through the centre. Diameter = 2× radius.
- An **arc** is a part of the circle. A chord cuts the circle into a **minor arc** and a **major arc**.
- A **segment** is the region between a chord and an arc.
- A **sector** is the region between two radii and an arc.



Note for the teacher. Ask students to draw a circle and name each part on their own drawing. Students often confuse a segment with a sector. A sector always has two radii as its sides; a segment has a chord.

2 Equal chords and angles at the centre (15 min)

Solve first.

P1. A circle has centre O and radius 5 cm. Chords AB and CD are each 6 cm long. Draw a rough diagram and join OA, OB, OC and OD.

(a) Write the three pairs of equal sides in $\triangle AOB$ and $\triangle COD$.

(b) Are the two triangles congruent? Name the rule.

(c) What can you say about $\angle AOB$ and $\angle COD$?

P2. In another circle with centre O and radius 5 cm, $\angle POQ = \angle ROS = 60^\circ$.

(a) What kind of triangle is $\triangle POQ$? Give a reason.

(b) Find the lengths of the chords PQ and RS.

(c) What do you notice about the two chords?

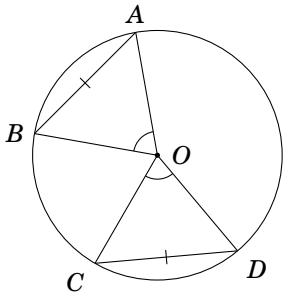
Answers.

P1. (a) $OA = OC = 5$ cm and $OB = OD = 5$ cm (radii); $AB = CD = 6$ cm (given). (b) Yes, $\triangle AOB \cong \triangle COD$ by SSS. (c) $\angle AOB = \angle COD$ (corresponding parts of congruent triangles).

P2. (a) $OP = OQ = 5$ cm, so $\angle OPQ = \angle OQP$. They add up to $180^\circ - 60^\circ = 120^\circ$, so each is 60°. The triangle is equilateral. (b) $PQ = 5$ cm and, in the same way, $RS = 5$ cm. (c) The chords are equal.

Note for the teacher. Let students work on P1 and P2 before you state Theorems 1 and 2. P1 leads to Theorem 1 and P2 leads to its converse.

Theorem 1. *Equal chords of a circle subtend equal angles at the centre.*

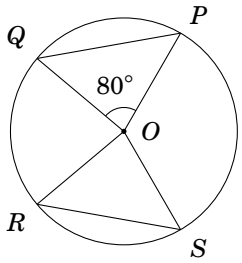


Why it works. In $\triangle AOB$ and $\triangle COD$: $OA = OC$ (radii), $OB = OD$ (radii) and $AB = CD$ (given). So $\triangle AOB \cong \triangle COD$ (SSS). Hence $\angle AOB = \angle COD$ (corresponding parts of congruent triangles).

Theorem 2 (Converse)**.** *If two chords of a circle subtend equal angles at the centre, then the chords are equal.*

Why it works. In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $\angle AOB = \angle COD$ and $OB = OD$. So $\triangle AOB \cong \triangle COD$ (SAS). Hence $AB = CD$.

Example 2.1. In a circle with centre O, chords PQ and RS are equal and $\angle POQ = 80^\circ$. Find $\angle ROS$ and $\angle ORS$.



Given: $PQ = RS$ and $\angle POQ = 80°$.
To find: $\angle ROS$ and $\angle ORS$.

1. $\angle ROS = \angle POQ = 80°$ (equal chords subtend equal angles at the centre, Theorem 1).
2. $OR = OS$ (radii), so $\triangle ORS$ is isosceles and $\angle ORS = \angle OSR$.
3. $\angle ORS + \angle OSR = 180° - 80° = 100°$ (angle sum of a triangle), so $\angle ORS = 50°$.

Answer: $\angle ROS = 80°$ and $\angle ORS = 50°$.

Note for the teacher. Stress that the angles are at the *centre*. Ask: "Would the result hold if the angles were measured at a point on the circle?" Leave the question open for a later lesson.

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3 The perpendicular from the centre to a chord (20 min)

Solve first.

P3. A circle has centre O and radius 41 cm. PQ is a chord, and $ON \perp PQ$ with $ON = 9$ cm. Draw a rough diagram and join OP and OQ.

(a) Name the two right triangles. Are they congruent? Name the rule.
(b) Find PN and NQ.
(c) What do you notice about the point N?

P4. A circle has centre O and radius 26 cm. Chord AB is 48 cm long, and M is the midpoint of AB. Join OA, OB and OM.

(a) Show that $\triangle OMA \cong \triangle OMB$. Name the rule.
(b) $\angle OMA + \angle OMB = ?$ Use this to find $\angle OMA$.
(c) Find the length OM.

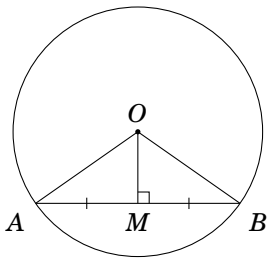
Answers.

P3. (a) $\triangle ONP$ and $\triangle ONQ$; $\angle ONP = \angle ONQ = 90°$, $OP = OQ$ (radii), ON common, so they are congruent by RHS. (b) $PN^2 = 41^2 - 9^2 = 1681 - 81 = 1600$, so $PN = 40$ cm, and $NQ = 40$ cm. (c) N is the midpoint of PQ.

P4. (a) $AM = BM$ (given), $OA = OB$ (radii), OM common, so $\triangle OMA \cong \triangle OMB$ by SSS. (b) $180°$ (linear pair). The two angles are equal, so $\angle OMA = 90°$. (c) $AM = 24$ cm, $OM^2 = 26^2 - 24^2 = 676 - 576 = 100$, so $OM = 10$ cm.

Note for the teacher. P3 leads to Theorem 3 and P4 leads to Theorem 4. Part (c) of each problem previews the calculation in the examples below.

Theorem 3. *The perpendicular from the centre of a circle to a chord bisects the chord.*



Why it works. Let $OM \perp AB$. In $\triangle OMA$ and $\triangle OMB$: $\angle OMA = \angle OMB = 90°$, $OA = OB$ (radii) and OM is common. So $\triangle OMA \cong \triangle OMB$ (RHS). Hence $AM = BM$.

Theorem 4 (Converse)**.** *The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.*

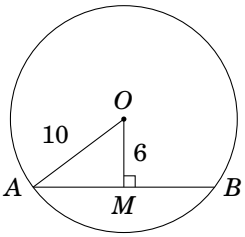
Why it works. Let M be the midpoint of AB. In $\triangle OMA$ and $\triangle OMB$: $AM = BM$, $OA = OB$ (radii) and OM is common. So $\triangle OMA \cong \triangle OMB$ (SSS) and $\angle OMA = \angle OMB$. These two angles form a linear pair, so each is $90°$.

Things to Remember

The **distance of a chord from the centre** is the length of the perpendicular from the centre to the chord. If the radius is r, the chord has length c and its distance from the centre is d, then the right triangle OMA gives

$$\left(\frac{c}{2}\right)^2 + d^2 = r^2.$$

Example 3.1. A chord of a circle of radius 10 cm is 6 cm from the centre. Find the length of the chord.

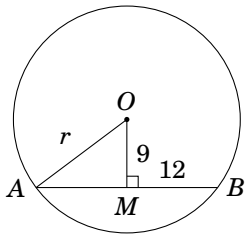


Given: radius $OA = 10$ cm, $OM \perp AB$, $OM = 6$ cm.
To find: the length of chord AB.

1. $AM = MB$ (the perpendicular from the centre bisects the chord, Theorem 3).
2. $\triangle OMA$ is right-angled at M, so $AM^2 = OA^2 - OM^2 = 100 - 36 = 64$ (Pythagoras theorem). So $AM = 8$ cm.
3. $AB = 2 \times AM = 2 \times 8 = 16$ cm.

Answer: the chord is 16 cm long.

Example 3.2. A chord of length 24 cm is 9 cm from the centre of a circle. Find the radius of the circle.



Given: $AB = 24$ cm, $OM \perp AB$, $OM = 9$ cm.
To find: the radius OA.

1. $AM = \dfrac{24}{2} = 12$ cm (Theorem 3).
2. $OA^2 = AM^2 + OM^2 = 144 + 81 = 225$ (Pythagoras theorem in $\triangle OMA$). So $OA = 15$ cm.

Answer: the radius is 15 cm.

Note for the teacher. Students often use the whole chord in place of half of it. Ask them to mark the midpoint M and the right angle before they calculate.

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4 Practice (15 min)

Work in your exercise book. Draw a rough diagram for every question.

Exercise 1A

1. The radius of a circle is 13 cm. A chord is 5 cm from the centre. Find the length of the chord. Draw a rough diagram first.
2. A chord of length 30 cm is drawn in a circle of radius 17 cm. Find the distance of the chord from the centre.
3. Chord AB of a circle with centre O is 48 cm long. $OM \perp AB$ and $OM = 7$ cm. Ravi writes: radius $= \sqrt{48^2 + 7^2}$. Find his mistake, and find the correct radius.
4. In a circle with centre O, chords AB and CD are equal and $\angle AOB = 70^\circ$. Find $\angle COD$ and $\angle OCD$. Give a reason for each step.

5. Draw a circle of radius 5 cm and a chord of length 8 cm in it. Use a set square to draw the perpendicular from the centre to the chord. Measure its length. Then find the length by calculation and compare.

6. A round dining table top has radius 50 cm. A carpenter fixes a straight wooden strip on the underside along a chord, 30 cm from the centre. Find the length of the strip.

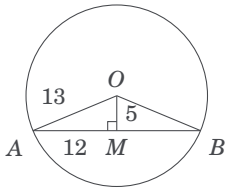
Note for the teacher. If time is short, set Questions 5 and 6 as homework. Questions 1 to 3 give the main practice in the relation $\left(\frac{c}{2}\right)^2 + d^2 = r^2$.

Answers.

1. $OM \perp AB$, so M is the midpoint of AB (Theorem 3). In right triangle OMA,

$$AM^2 = OA^2 - OM^2 = 13^2 - 5^2 = 169 - 25 = 144,$$

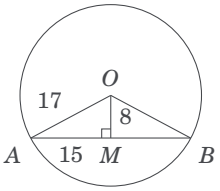
so $AM = 12$ cm, and $AB = 2 \times 12 = 24$ cm. **Answer:** 24 cm.



2. $AM = \frac{30}{2} = 15$ cm. In right triangle OMA,

$$OM^2 = OA^2 - AM^2 = 17^2 - 15^2 = 289 - 225 = 64,$$

so $OM = 8$ cm. **Answer:** 8 cm.



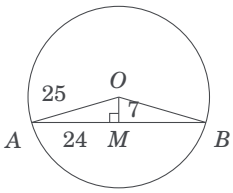
3. Ravi used the whole chord in place of the half chord. M is the midpoint of AB, so

$$AM = \frac{48}{2} = 24 \text{ cm}.$$

In right triangle OMA,

$$OA^2 = AM^2 + OM^2 = 24^2 + 7^2 = 576 + 49 = 625,$$

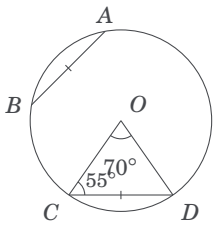
so $OA = 25$ cm. **Answer:** the radius is 25 cm.



4. Equal chords subtend equal angles at the centre, so $\angle COD = \angle AOB = 70°$ (Theorem 1). In triangle OCD, $OC = OD$ because both are radii, so the triangle is isosceles and $\angle OCD = \angle ODC$. Therefore

$$\angle OCD = \frac{180° - 70°}{2} = \frac{110°}{2} = 55°.$$

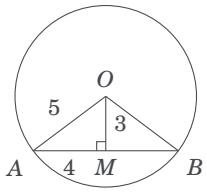
Answer: $\angle COD = 70°$, $\angle OCD = 55°$.



5. The half chord is $AM = 4$ cm. In right triangle OMA,

$$OM^2 = OA^2 - AM^2 = 5^2 - 4^2 = 25 - 16 = 9,$$

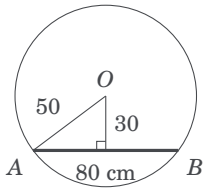
so $OM = 3$ cm. The measured length should be close to 3 cm. **Answer:** 3 cm.



6. The strip is a chord. The half chord is

$$\sqrt{50^2 - 30^2} = \sqrt{2500 - 900} = \sqrt{1600} = 40 \text{ cm},$$

so the strip is $2 \times 40 = 80$ cm long. **Answer:** 80 cm.



5 Exit question (5 min)

Answer on a slip of paper.

(a) True or false? "Every line through the midpoint of a chord passes through the centre." Give a reason.

(b) A chord of length 10 cm is drawn in a circle of radius 13 cm. How far is the chord from the centre?

Answers.

(a) False. The line joining the centre to the midpoint of a chord is perpendicular to the chord (Theorem 4). A slanted line through the midpoint does not pass through the centre. Only the perpendicular from the centre bisects the chord (Theorem 3). **Answer:** false.

(b) The half chord is 5 cm. In right triangle OMA,

$$d^2 = 13^2 - 5^2 = 169 - 25 = 144,$$

so $d = 12$ cm. **Answer:** 12 cm.

